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Title: Sustainable Building Materials in Africa (2015–2025): A Rapid Systematic Review

Background and Rationale

The construction industry is among the largest global contributors to resource depletion and greenhouse gas emissions. Sustainable building materials (SBMs) are increasingly promoted to reduce environmental burdens while improving durability, affordability, and socio-economic value.

In Africa, rapid urbanization and housing demand have placed immense pressure on the construction sector. While there is growing recognition of sustainability, the adoption of SBMs across the continent remains fragmented, with research dispersed across countries, material types, and contexts. To date, no consolidated review has synthesized evidence on sustainable materials specific to Africa within the past decade.

This review seeks to address that gap by systematically identifying SBMs studied in Africa, their reported benefits, adoption challenges, and implications for policy and practice.

Objectives

- To identify the range of sustainable building materials researched or applied in Africa between 2015–2025.
- To classify these materials by sustainability dimensions: environmental, economic, social, and technical.
- To evaluate reported performance indicators and benefits of SBMs.
- To analyze barriers to adoption and highlight enabling factors.
- To provide recommendations for construction professionals, policymakers, and researchers.

Eligibility Criteria

Inclusion Criteria

- Peer-reviewed journal articles, conference proceedings, and reputable institutional/NGO reports.
- Publications between January 2015 and December 2025.
- Studies explicitly addressing sustainable building materials in African countries.
- Articles written in English.

Exclusion Criteria

1. Studies conducted outside Africa.
2. Non-English publications.
3. Papers on general sustainability without material focus.
4. Editorials, opinion pieces, or non-empirical reports.

Databases and Search Strategy

Databases to Search:

1. Scopus
2. Web of Science
3. ScienceDirect
4. African Journals Online (AJOL)
5. Google Scholar (for supplementary coverage)

Search String (adapted per database):

- (*"sustainable building materials" OR "green construction materials" OR "eco-friendly materials"*)
- *AND ("Africa" OR "African countries")*
- *AND ("construction" OR "building")*
- *AND (2015:2025[Date - Publication])*

Supplementary Search:

Reference snowballing will be conducted on included studies to capture additional relevant literature.

Screening and Selection Process

- Titles and abstracts will be screened for relevance.
- Full-text screening will follow, based on eligibility criteria.
- The **PRISMA 2020 flow diagram** will document the number of studies identified, screened, excluded, and included.
- Screening will be conducted by the primary reviewer, with a sample double-checked for consistency.

Data Extraction

A structured Excel sheet will be used. Extracted data will include:

1. Author(s), Year
2. Country / Region
3. Study Type (case study, experimental, survey, review, etc.)
4. Material(s) Investigated
5. Application Context (housing, commercial, infrastructure, lab)
6. Sustainability Dimension(s) (environmental, economic, social, technical)
7. Performance Indicators (e.g., strength, durability, emissions, cost)
8. Reported Benefits
9. Reported Barriers / Challenges
10. Adoption Stage (conceptual, pilot, implemented)
11. Stakeholder Perspective (industry, academic, policy, mixed)
12. Policy/Regulatory Mention
13. Funding/Support Mentioned
14. Recommendations
15. Quality Rating (High / Medium / Low, per JBI)

Quality Assessment

The **Joanna Briggs Institute (JBI) critical appraisal checklists** will be applied according to study type (qualitative, cross-sectional, randomized trial, etc.).

Studies will be rated as:

- **High quality** → meets most criteria, minimal bias.
- **Medium quality** → some limitations, moderate bias.
- **Low quality** → fails multiple criteria, high risk of bias.

Quality ratings will inform interpretation but not exclude studies.

Data Synthesis and Analysis

1. A **thematic synthesis** approach will be employed.
2. Materials will be categorized (e.g., natural-based, recycled/reused, innovative composites).
3. Reported barriers and benefits will be grouped by sustainability dimensions.
4. Frequency counts will highlight commonly cited materials, benefits, and barriers.
5. Cross-country comparison will reveal regional differences and policy gaps.
6. A narrative synthesis will contextualize findings and outline implications.

Registration and Dissemination

1. This protocol will be registered on the **Open Science Framework (OSF)** for transparency and timestamping.
2. The review will be submitted to a peer-reviewed journal focused on construction, sustainability, or African development.
3. Findings will also be disseminated through conferences and targeted policy briefs.

Timeline (Rapid Review: 4 Weeks)

1. **Week 1:** Finalize protocol, register on OSF, design search strings, conduct database searches.
2. **Week 2:** Title/abstract screening, full-text screening.
3. **Week 3:** Data extraction and quality assessment.
4. **Week 4:** Data synthesis, write-up, and prepare manuscript for submission.